

Inkspire: Sketching Product Designs with AI

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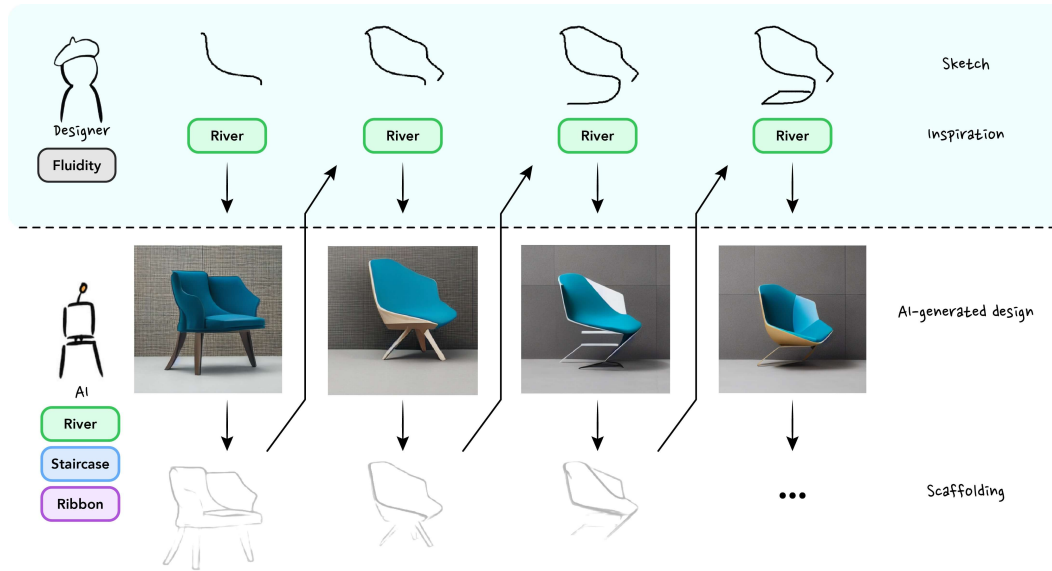


Figure 1: Inkspire supports an iterative sketching process between the designer and AI. The designer uses abstract concepts and loose sketch strokes and the AI suggests analogical inspirations and generated designs. In addition, the AI converts designs back into sketches as underlays of the designer's canvas (Figure 5). This helps the designer get inspired and build on AI designs.

ACM Reference Format:

David Chuan-En Lin, Hyeonsu B. Kang, Nikolas Martelaro, Aniket Kittur, Yan-Ying Chen, and Matthew K. Hong. 2024. Inkspire: Sketching Product Designs with AI. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3672539.3686339>

1 INTRODUCTION

Generative AI models are now able to synthesize ultra-realistic images using text [10, 13]. However, their integration into the designer's creative process is not without its challenges. We conducted

a comprehensive day-long exchange session with designers in a major automotive company. Designers showed researchers their design processes with examples from their work and showed their process of using current T2I tools, and we discussed new ideas on how to design tools to support them. From the session, we identified three key challenges: (1) Designers find it unnatural to design with AI primarily through language-based prompts. (2) Designers struggle to create inspiring designs with abstract language, typical in design briefs, using AI models. (3) Designers find designs generated by AI models “too complete” and challenging to build on. To address these challenges, we created Inkspire to help designers create product designs with AI through their natural way of designing: using abstract language and loose and iterative sketch strokes. Inkspire helps designers generate designs through sketching and also *build on generated designs through sketching* — we think this closed feedback loop facilitates a different kind of interactive sketching workflow with AI.

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UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

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ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686339>

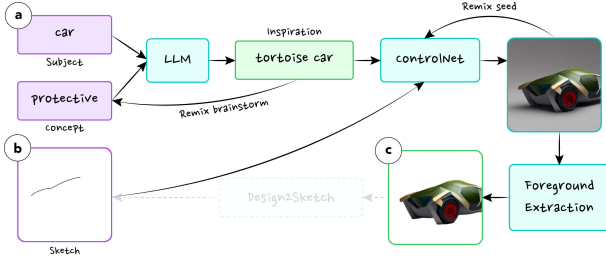


Figure 2: Sketch2Design pipeline: (a) concept generation, (b) sketch-guided generation, and (c) foreground extraction.

2 INKSPIRE

The Inkspire system consists of three main components: *Sketch2Design*, *Design2Sketch*, and *Design2Real*.

2.1 Sketch2Design

The *Sketch2Design* component helps users brainstorm design concepts and generate product designs through sketching (Figure 2). First, the user specifies the subject that they are designing for (e.g., car) and an initial abstract concept (e.g., protective). To brainstorm more concrete design ideas for the abstract concept, we leverage Large Language Models (LLMs) [2] to generate analogical inspirations (Figure 2a). Prior work has found that breaking down a problem into steps can improve the correctness of the LLM [14]. We thus take a two-step approach. We first prompt the LLM to detail the design principles for the given subject, which serves as context [9] for the LLM to generate more well-thought-out inspirations in Step 2. Given the design principles, we prompt the LLM to generate inspirations from the domains of nature, architecture, and fashion.

After selecting a design inspiration, the user may create product designs by sketching on a canvas. The user may start with as little as a single stroke (Figure 2b). Using ControlNet [16] to guide Stable Diffusion [13], we generate a product design guided by the stroke. We initialize the model with a low guidance scale (parameter for controlling how closely the AI adheres to user input) to handle incomplete sketches (which does not work well with the default guidance scale), then gradually increase it for precise control with more complete sketches. Each time a stroke is drawn, we generate a new design, making the creation process iterative and implicitly encouraging users to focus on sketching instead of on engineering text prompts (i.e., the current paradigm of working with T2I models). We maintain the same initial seed between generations to maintain consistency and for speedy generations. The user may click the “remix” button to change to a different seed and generate varied designs. Finally, we remove redundant backgrounds from the generated design using foreground extraction [11] (Figure 2c).

2.2 Design2Sketch

The *Design2Sketch* component helps users build on top of previously generated designs by converting them into sketches (Figure 3). Given the converted sketch, the user may visualize it as an underlay on their canvas, similar to tracing paper [1]. This enables the user to draw inspiration from aspects of the previously generated designs and also helps them overcome the challenge of starting with a blank

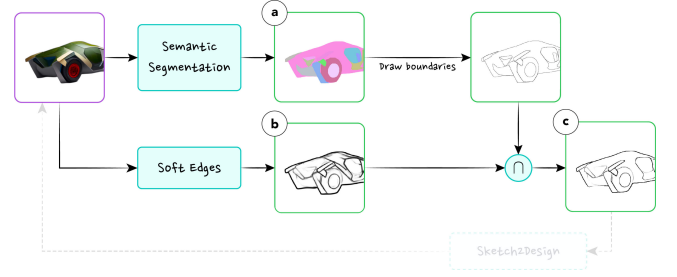


Figure 3: Design2Sketch pipeline: (a) semantic segmentation, (b) soft edge extraction, (c) computing intersection.

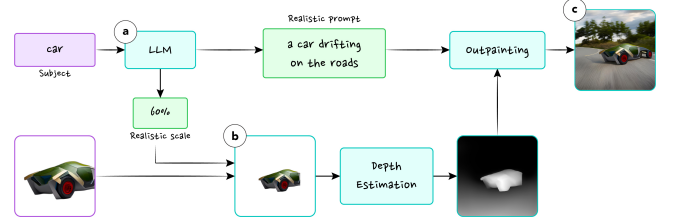


Figure 4: Design2Real pipeline, including (a) realistic prompt and scale generation, (b) depth estimation, (c) outpainting.

canvas [6], especially during the early stages of sketching. The user can continue iterating through sketching, completing the sketch-to-design-to-sketch feedback loop. We introduce a novel approach for converting designs to sketches. First, we perform semantic segmentation on the design [7]. Given the segmentation map, we then draw the boundaries (Figure 3a). Second, we extract soft edges from the design [15] (Figure 3b). Finally, we take the intersection between the segmentation map boundaries and the extracted soft edges as the sketch (Figure 3c). This approach combines the best of both worlds from segmentation and edge extraction: It focuses on the design’s main structural lines and eliminates the redundant lines caused by texture (a limitation of edge extraction methods [4, 15]) while creating a sketch look with varying thickness of lines.

2.3 Design2Real

The *Design2Real* component helps users visualize how a generated design would look in a realistic environment (Figure 4). The component includes a pipeline of LLM prompting, rescaling, depth estimation, and outpainting. First, we prompt the LLM to describe the subject in a realistic scenario (Figure 4a). We also prompt the LLM to reason an appropriate scale for the main subject and rescale the design (from 30% to 100%) (Figure 4b). Second, we generate a depth map of the design with a depth estimation method [12]. This serves as an approximation of the design’s 3D structure. Finally, given the depth map and the prompt describing the subject in a realistic scenario, we outpaint the image (Figure 4c). Given the 3D approximation prior from the depth map, we are able to plant the design into a scene with realistic surface lighting and shadows.

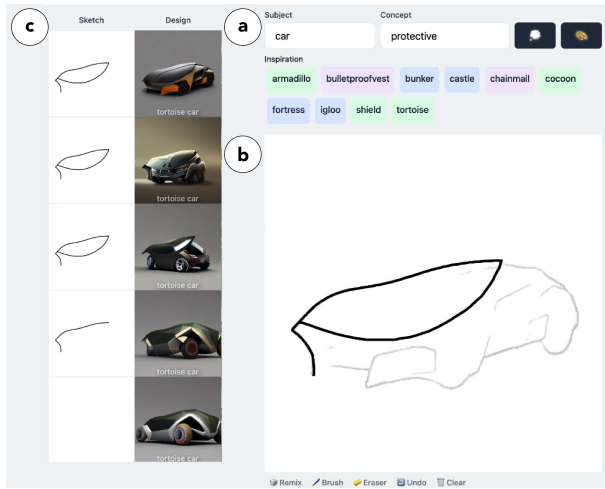


Figure 5: Inkspire interface, including (a) the Analogical Panel, (b) the Sketching Panel, and (c) the Evolution Panel.

3 USER STUDY

We conducted a within-subjects user study with twelve participants (half expert designers and half novices) comparing Inkspire (excluding the Design2Real component) with a baseline of a typical ControlNet setup [16]. Designers were asked to design a lamp or a chair using either of the two conditions (counterbalanced) through an hour-long study. Overall, we found that Inkspire increased back-and-forth between designer and AI through divergent and convergent mechanisms (via interaction data, Figures 6 and 7), decreased manual prompting and more semantically diverse prompts (via prompt BERTScores), increased sketching frequency and higher perceived “sense of sketching” with a decreased sunk cost effect (via ratings on Buxton’s sketching principles [3], Figure 8), increased support of a creative workflow (via CSI questions [5], Figure 9), and increased ability to get a sense of the current state of the AI through live scaffolding and evolution of designs (via Human-AI Collaboration questions [8], Figure 10).

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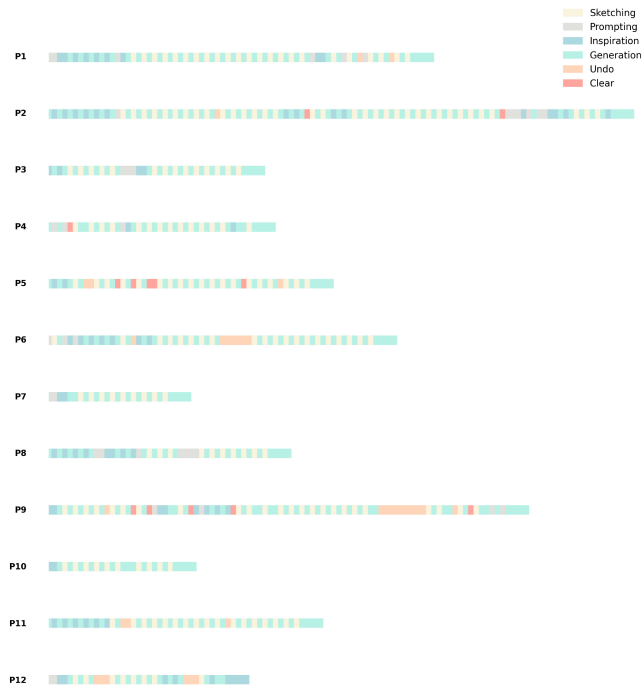


Figure 6: Participant usage logs when using Inkspire.

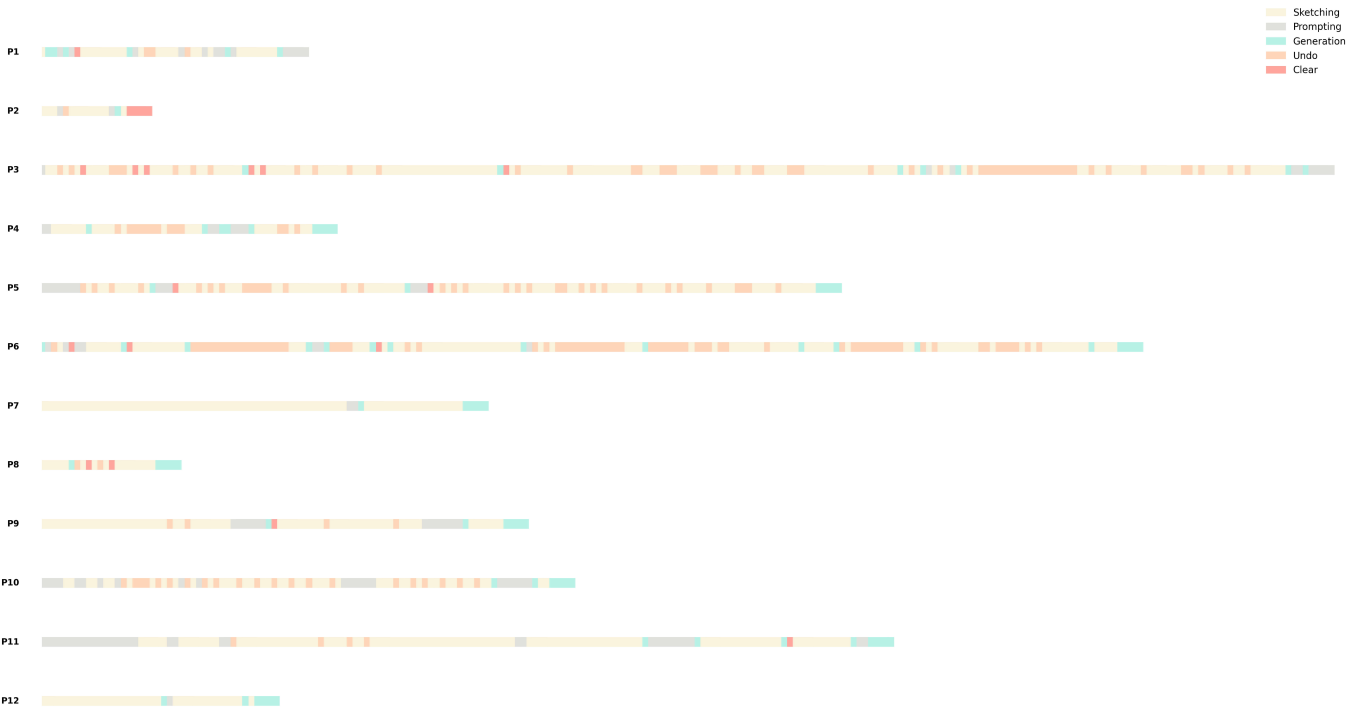


Figure 7: Participant usage logs when using the baseline condition.

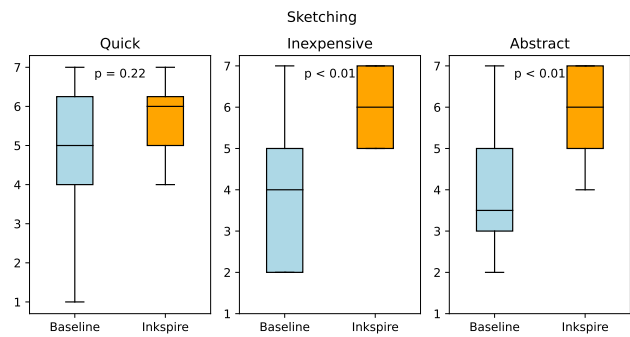


Figure 8: Results on sketching experience, including how sketches feel quick, inexpensive, and abstract (7-point Likert scale, higher is better).

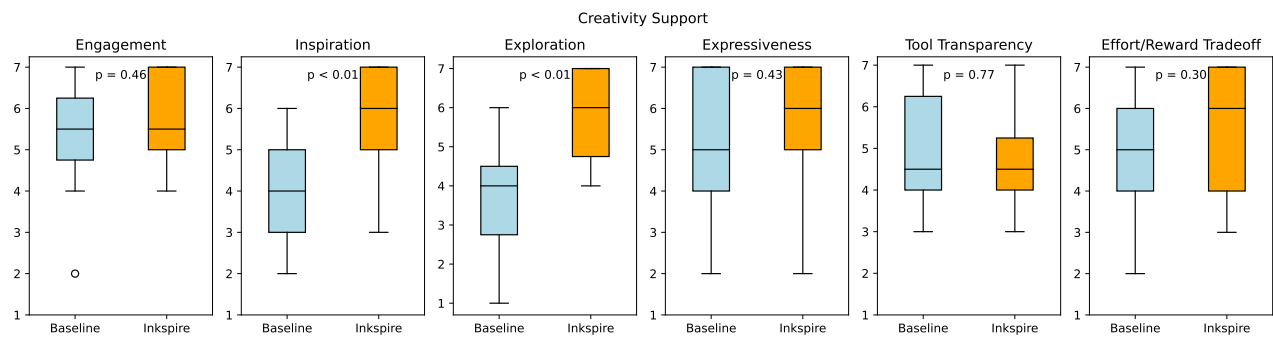


Figure 9: Results on creativity measured with the Creativity Support Index (CSI) [5] (7-point Likert scale, higher is better).

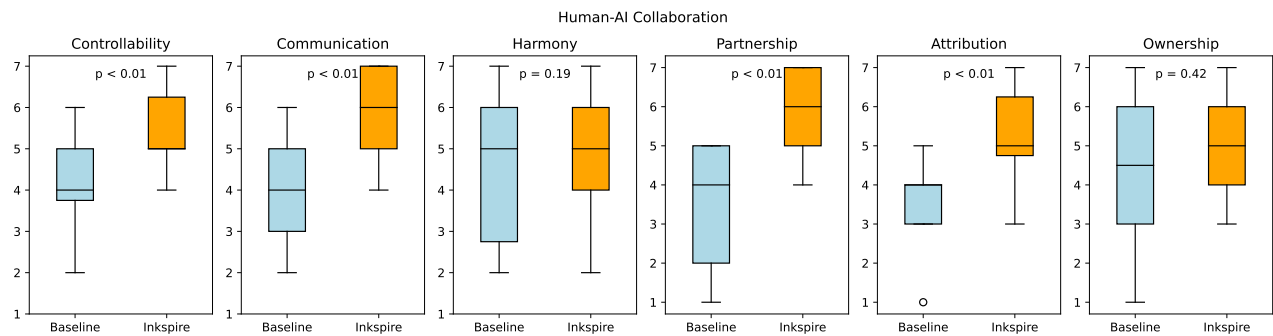
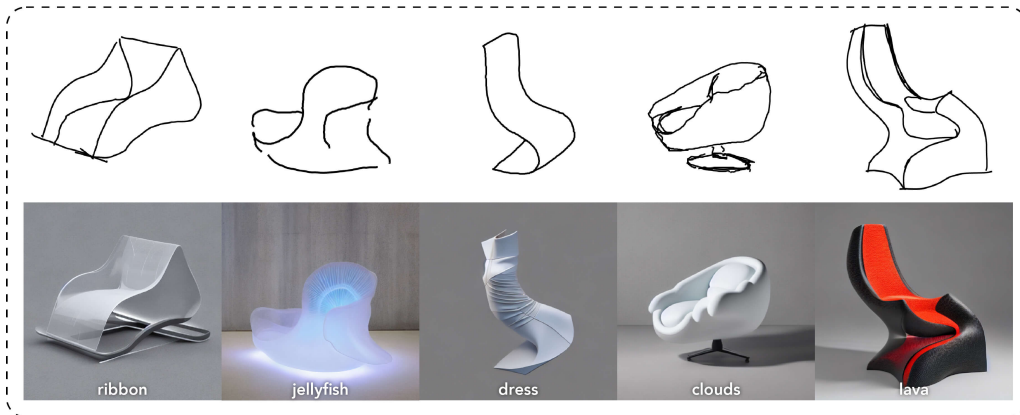


Figure 10: Results on human-AI collaboration measured with questions from [8] (7-point Likert scale, higher is better).

"fluid chair"



"serene lamp"

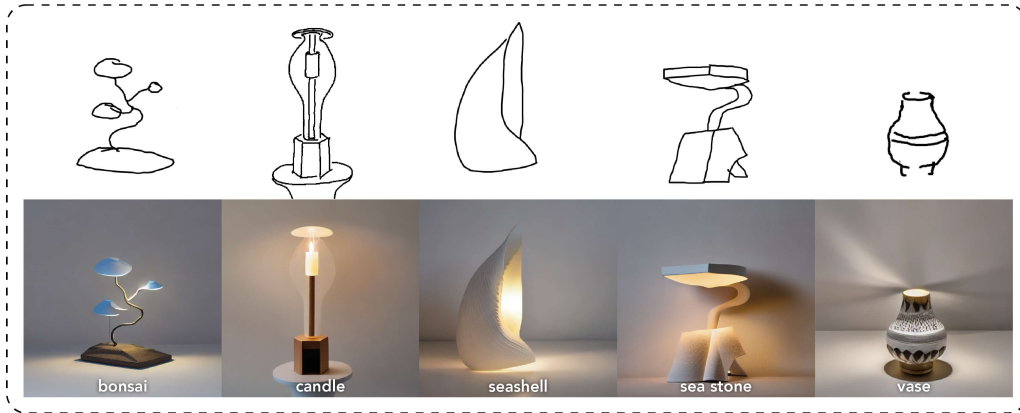


Figure 11: Example designs created by users using Inkspire. The final user sketch is shown on top and the generated image is shown on the bottom, along with the selected analogy word chosen by the user.